

Multi-order Markov Chain for Next-Node Prediction with Long and Short Term Memories

Cappu & Matcha & Karlitos

May 28, 2026

1 Brainstorming

1. ✓ Use WorldMove trajectories data
2. LTM trained on all trajectories of all users (population model) and STM feed with a single user trajectory (individual model) ?
3. How node are encoded in the other papers ?
4. Compare to lower/higher order (result for different orders)
5. Compare to AKOM, CPT+, MOGen, IOHMM
6. Explainability analysis
7. compare to probabilistic model of the network based on previous trajectories (combined all prediction)
8. Use ideas of Predictive information as framework (check if the past of the user is enough or if we need the past of the whole system)
9. Including recency decay over used edges (STM/LTM)
10. STM train on one day of one user / LTM train on 7 seven dat of all users (experts=similar users)
11. Data with Activity on multiple days
12. LTM cluster by CL/CI/CO (found them) and compare with real trajectories

2 Related work

This section reviews the existing literature relevant to our proposed approach: multi-order Markov chain models for next-node prediction in road networks, with a dual Long-Term Memory (LTM) and Short-Term Memory (STM) architecture inspired by the IDyOM model [8]. We organise the review around four axes: (i) higher-order and multi-order Markov models applied to network paths, (ii) their specific application to road and transit networks, (iii) hybrid population-level and individual-level prediction approaches, and (iv) the original IDyOM framework and its LTM/STM combination mechanism.

2.1 Higher-Order and Multi-Order Markov Models on Network Paths

A classical limitation of first-order Markov chains for trajectory modelling is the memoryless assumption: the next node depends only on the current node, ignoring all prior context. Higher-order Markov chains address this by conditioning on the last k visited nodes, but selecting the appropriate order k is non-trivial and context-dependent.

Scholtes [12] introduced the *Multi-Order Graphical Model* (MOM), which selects the appropriate Markov order for path data in temporal networks via a likelihood-based model selection criterion. This laid the theoretical groundwork for combining Markov chains of several orders into a single representation.

Building on this, Gote et al. [3] proposed *MOGen*, a generative multi-order modelling framework that combines Markov chains of multiple orders into a single model. MOGen accounts for sequential patterns in traversed node sequences that are constrained by an underlying network topology, and includes an information-theoretic model selection algorithm to automatically determine the optimal maximum order directly from data. Applied to passenger itineraries in the London Underground and several clickstream datasets, MOGen outperforms state-of-the-art sequence modelling baselines including AKOM [10] and CPT+ [4]. While MOGen is conceptually close to our proposal in its multi-order treatment of network-constrained paths, it trains a single unified model on all available trajectory data and does not distinguish between a population-level long-term memory and an individual- or session-level short-term memory.

Singer et al. [13] and West and Leskovec [14] showed that non-Markovian correlations in navigation sequences, correlations not captured by the topology alone, significantly improve prediction performance when modelled explicitly, motivating the use of higher-order representations beyond the simple graph structure.

2.2 Markov-Based Next-Node Prediction in Road and Transit Networks

The application of Markov chain models to road and transit network prediction has a long history. Standard approaches model trajectories as first- or

second-order Markov chains over road segments or intersections, where transition probabilities are estimated from historical GPS or smart card data [2].

More recently, Zhang et al. [16] proposed a higher-order network framework specifically for transportation systems, extending classical graph representations using higher-order Markov chains and de Bruijn graph structures. Their framework generalises betweenness centrality, PageRank, and next-step prediction to this higher-order setting, and validates the approach on the Sioux Falls network using agent-based trajectory data. Experimental results show that higher-order models consistently outperform first-order baselines, with the third-order model achieving an optimal trade-off between predictive accuracy and model complexity. Like MOGen, however, this approach trains a single model on all trajectories and offers no mechanism for online adaptation to the current trajectory.

For transit networks, Mo et al. [7] proposed an Input-Output Hidden Markov Model (IOHMM) to jointly predict the next trip location and timing of individual metro passengers. The model incorporates hidden activity patterns and achieves comparable performance to LSTM-based approaches while offering interpretable behavioural explanations. Although the IOHMM implicitly captures individual-level regularities through its hidden state structure, it does not explicitly combine a population-level prior model with an online individual model.

Deep learning approaches, including LSTM models [1, 6] and Transformer architectures [15], have demonstrated strong performance on next-node and next-segment prediction tasks by learning long-range temporal dependencies. However, these models are generally trained offline on the full trajectory dataset and lack an explicit mechanism for fast, lightweight adaptation to the ongoing trajectory of a specific vehicle or user.

2.3 Combining Population-Level and Individual-Level Models

A key motivation for a dual LTM/STM architecture is the complementarity between general mobility patterns shared across all users and patterns specific to an individual’s recent history.

Qiao et al. [11] proposed a hybrid variable-order Markov model for human mobility prediction that first mines frequent mobility sequences at the individual level to determine the appropriate Markov order per user, and then incorporates a temporal factor into the predictor. While this approach adapts to individual behaviour, it does so using only each user’s own historical data and does not exploit a shared population-level model.

Hawelka et al. [5] addressed the cold-start problem for users with short trajectory histories by combining a large, diverse corpus of mobility sequences from multiple users, a population-level model, with individual sequence predictors. Their experiments on roaming mobile phone data showed that the collective predictor substantially outperforms individual Markov models for users with short histories, demonstrating the complementary value of population-level and individual-level information. This population/individual duality is conceptually analogous to the LTM/STM split in IDyOM, but the combination mechanism

is different: no variable-order Markov modelling or entropy-based weighting is used.

2.4 The IDyOM Framework: LTM, STM, and Entropy-Weighted Combination

The Information Dynamics of Music model (IDyOM) [8, 9] is a computational model of melodic expectation grounded in probabilistic prediction via variable-order Markov chains. IDyOM maintains two complementary sub-models. The *Long-Term Model* (LTM) is trained offline on a large corpus of melodies and captures statistical regularities shared across the musical culture. The *Short-Term Model* (STM) is trained online and incrementally on the current melody as it unfolds, capturing patterns specific to that particular piece. At each prediction step, the probability distributions produced by the LTM and STM are combined via a geometric mean weighted inversely by their respective entropies: a model that is more uncertain (higher entropy) about the next event contributes less to the combined prediction [9]. This entropy-weighted combination allows IDyOM to rely on the population-level prior when individual context is scarce, and to rely increasingly on local patterns as the current sequence provides more information.

To the best of our knowledge, the IDyOM architecture, a multi-order Markov LTM trained on the full trajectory corpus combined with a dynamically updated multi-order Markov STM trained on the current or recent trajectory, merged via entropy-weighted combination, has not been transposed to the road network trajectory prediction setting. The present work addresses this gap by adapting the IDyOM dual-memory paradigm to the problem of next-node prediction in road networks, where the LTM captures population-level driving patterns and the STM adapts online to the individual trajectory unfolding in real time.

2.5 Summary and Positioning

Table 1 summarises the main related approaches along the dimensions most relevant to our contribution. As shown in this Table, no existing work combines all four properties simultaneously: multi-order Markov modelling, application to road or transit networks, an explicit population-level Long-Term Model, and an explicit session-level Short-Term Model with entropy-weighted combination. Our approach is the first, to our knowledge, to bring this IDyOM-inspired dual-memory paradigm to the road network next-node prediction problem.

3 Model

3.1 Encoding of the nodes

Across the reviewed approaches, a trajectory of length T is represented as an ordered sequence $\mathbf{s} = (v_1, v_2, \dots, v_T)$, where each element v_t belongs to a node

Table 1: Comparison of related approaches for next-node prediction in networks.

Model	Multi-order Markov	Road/transit network	Explicit LTM	Explicit STM
gote2023mogen	✓	✓	–	–
Zhang2025higherorder	✓	✓	–	–
qiao2018hybrid	✓	–	–	✓
Hawelka2017collective	–	–	✓	–
mo2021individual	–	✓	–	–
pearce2005idyom	✓	–	✓	✓
Proposed	✓	✓	✓	✓

alphabet $\mathcal{V}$. The choice of how to define $\mathcal{V}$ and encode each v_t varies significantly between methods.

Categorical ID encoding. The simplest and most common encoding, used by MOGen [3], the hybrid Markov model [11], and IOHMM [7], maps each node to a discrete integer identifier: $v_t \in \{1, \dots, N\}$ where $N = |\mathcal{V}|$. A k -th order Markov chain then models the conditional distribution

$$P(v_{t+1} \mid v_{t-k+1}, \dots, v_t), \quad (1)$$

with feasible transitions constrained by a binary adjacency matrix $A \in \{0, 1\}^{N \times N}$ encoding the road topology, i.e. $P(v_{t+1} \mid \cdot) = 0$ whenever $A[v_t, v_{t+1}] = 0$.

Higher-order (de Bruijn) node encoding. Zhang et al. [16] lift the alphabet to k -grams of base nodes. A order- k node is a tuple $\tilde{v} = (v_{t-k+1}, \dots, v_t) \in \mathcal{V}^k$, yielding an expanded state space of size up to N^k . The transition model becomes

$$P((v_{t-k+2}, \dots, v_{t+1}) \mid (v_{t-k+1}, \dots, v_t)), \quad (2)$$

which is equivalent to a first-order Markov chain on the de Bruijn graph $\mathcal{G}^{(k)}$ whose nodes are valid k -grams.

Graph-embedding encoding. NetTraj [6] represents each intersection v_t as a continuous vector $\mathbf{e}_t \in \mathbb{R}^d$ obtained via Node2Vec, augmented with a discretised bearing angle $\theta_t \in [0, 2\pi)$:

$$\phi(v_t) = [\mathbf{e}_t \parallel \theta_t] \in \mathbb{R}^{d+1}, \quad (3)$$

where $\parallel$ denotes concatenation. This embedding is then fed into an LSTM encoder, replacing the explicit Markov transition table with learned recurrent dynamics.

References

- [1] F. Althé and A. de La Fortelle. An lstm network for highway trajectory prediction. 2018.

- [2] M. Chen, X. Yu, and Y. Liu. Mining moving patterns for predicting next location. *Information Systems*, 54:156–168, 2015.
- [3] C. Gote, G. Casiraghi, F. Schweitzer, and I. Scholtes. Predicting variable-length paths in networked systems using multi-order generative models. *Applied Network Science*, 8(1):68, 2023.
- [4] T. Gueniche, P. Fournier-Viger, and V. S. Tseng. Compact prediction tree: A lossless model for accurate sequence prediction. In H. Motoda, Z. Wu, L. Cao, O. Zaiane, M. Yao, and W. Wang, editors, *Advanced Data Mining and Applications*, pages 177–188, Berlin, Heidelberg, 2013. Springer Berlin Heidelberg.
- [5] B. Hawelka, I. Sitko, P. Kazakopoulos, and E. Beinat. Collective prediction of individual mobility traces for users with short data history. *PLOS ONE*, 12(1):1–14, 01 2017.
- [6] Y. Liang and Z. Zhao. Nettraj: A network-based vehicle trajectory prediction model with directional representation and spatiotemporal attention mechanisms. *IEEE Transactions on Intelligent Transportation Systems*, 23(9):14470–14481, 2022.
- [7] B. Mo, Z. Zhao, H. N. Koutsopoulos, and J. Zhao. Individual mobility prediction in mass transit systems using smart card data: An interpretable activity-based hidden markov approach. *IEEE Transactions on Intelligent Transportation Systems*, 23(8):12014–12026, 2022.
- [8] M. T. Pearce. The construction and evaluation of statistical models of melodic structure in music perception and composition. 2005.
- [9] M. T. Pearce. Statistical learning and probabilistic prediction in music cognition: Mechanisms of stylistic enculturation. *Annals of the New York Academy of Sciences*, 1423(1):378–395, 2018.
- [10] J. Pitkow and P. Pirolli. Mining longest repeated subsequences to predict world wide web surfing. In *Second USENIX Symposium on Internet Technologies & Systems (USITS 99)*, Boulder, CO, Oct. 1999. USENIX Association.
- [11] Y. Qiao, Z. Si, Y. Zhang, F. B. Abdesslem, X. Zhang, and J. Yang. A hybrid markov-based model for human mobility prediction. *Neurocomputing*, 278:99–109, 2018. Recent Advances in Machine Learning for Non-Gaussian Data Processing.
- [12] I. Scholtes. When is a network a network? multi-order graphical model selection in pathways and temporal networks. page 1037–1046, 2017.
- [13] P. Singer, D. Helic, B. Taraghi, and M. Strohmaier. Detecting memory and structure in human navigation patterns using markov chain models of varying order. *PLOS ONE*, 9(7):1–21, 07 2014.

- [14] R. West and J. Leskovec. Human wayfinding in information networks. page 619–628, 2012.
- [15] S. Yang, J. Liu, and K. Zhao. Getnext: Trajectory flow map enhanced transformer for next poi recommendation. In *Proceedings of the 45th International ACM SIGIR Conference on Research and Development in Information Retrieval*, SIGIR '22, page 1144–1153, New York, NY, USA, 2022. Association for Computing Machinery.
- [16] C. Zhang and J. Hackl. Beyond connectivity: Higher-order network framework for capturing memory-driven mobility dynamics. 2025.